

PART #01

PYTHON PANDAS

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What is Pandas ?

- It is a library that provides easy to use data structure and data analysis tool for Python Programming Language.
- Pandas is used for data manipulation, analysis and cleaning.
- Tabular data with heterogeneously-typed columns.
- Mostly used in Machine Learning.



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installation

```
pip install pandas
```

Import Convention

```
import pandas as pd
```

Pandas Data Structure

1. Series

A series is a one-dimensional data structure.

2. DataFrame

A data frame is a tabular data, with rows to store the information and columns to name the information.



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Pandas Series

```
1 import pandas as pd
2 s = pd.Series([1., 2., 3.])
3 print(s)
```

```
0    1.0
1    2.0
2    3.0
dtype: float64
```

```
1 import pandas as pd
2 s1 = pd.Series([1, 2, 3], index=['a', 'b', 'c'])
3 print(s1)
```

```
a    1
b    2
c    3
dtype: int64
```



index



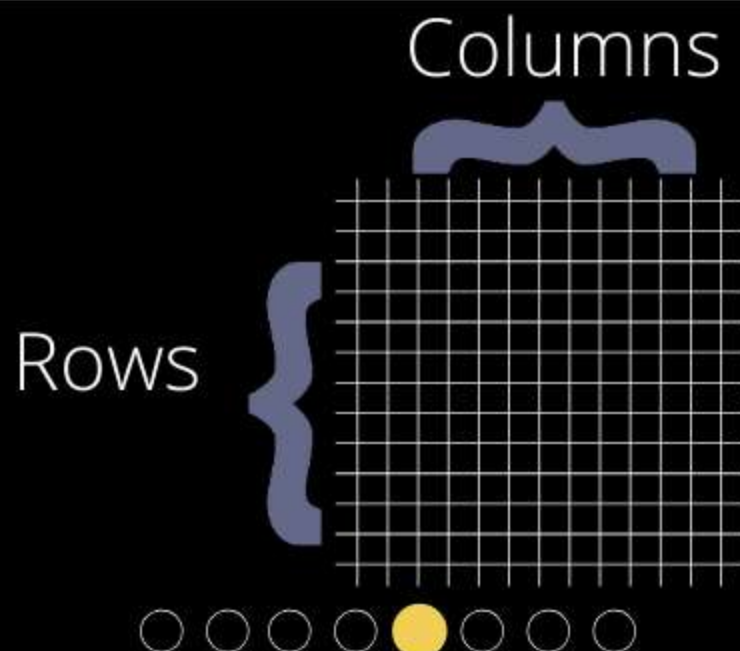
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Pandas Data Frame

Creating DataFrame using Dictionary

```
1 data_mobile = {'Mobile': ['iPhone', 'Samsung', 'Redmi'],  
2               'Color': ['Red', 'White', 'Black'],  
3               'Price': [5000, 4500, 4300]}  
4  
5 df = pd.DataFrame(data_mobile)  
6 print(df)
```

	Mobile	Color	Price
0	iPhone	Red	5000
1	Samsung	White	4500
2	Redmi	Black	4300



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Create Test/Fake Data

Creates a series from an iterable new_series

```
1 new_series = [1,2,3] #any iterable
2 s3 = pd.Series(new_series)
3 print(s3)
```



```
0    1
1    2
2    3
dtype: int64
```

String,
tuple,
list,
etc.



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Create Test/Fake Data

Create a DF of 3 columns and 4 rows of random floats

```
1 import numpy as np
2 df2 = pd.DataFrame(np.random.rand(4,3))
3 df2
```

	0	1	2
0	0.217182	0.815998	0.023604
1	0.617268	0.405259	0.472098
2	0.894960	0.178855	0.364088
3	0.594534	0.038284	0.779797

row

column



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